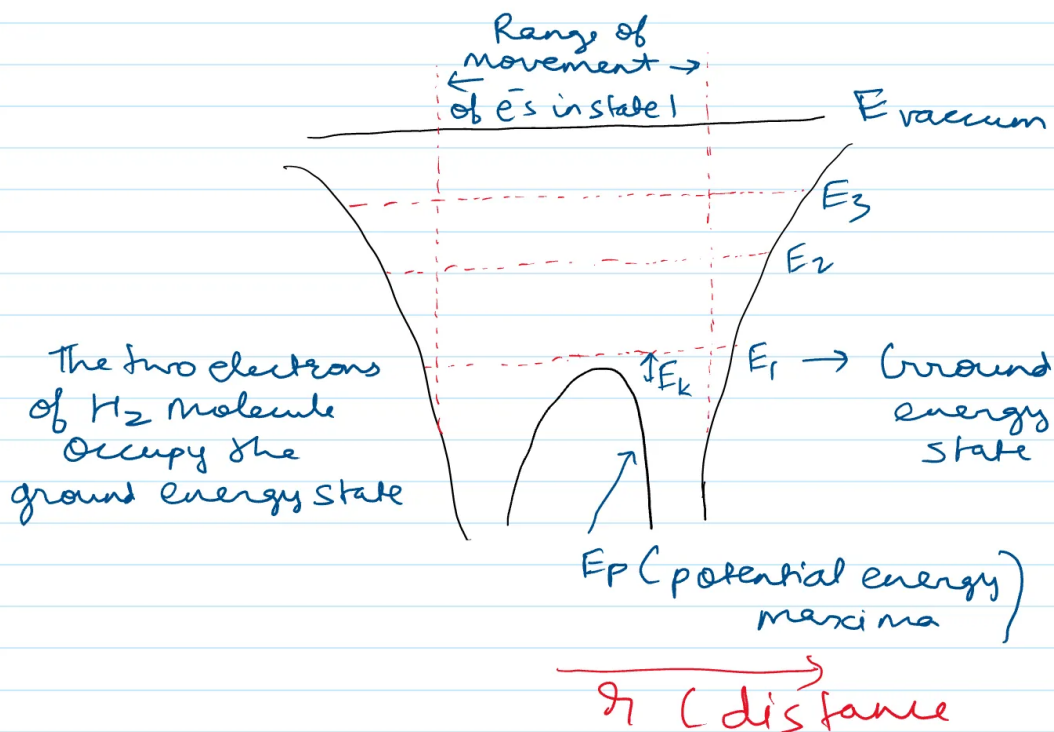


Quiz 2 Solution

- 1) For two hydrogen atoms, plot the potential energy and mention the status of allowed energy states with potential energy when both the atoms are very close, such that all the energy states are shared with both the atoms. What is the condition for stable co-valent bond formations? Describe using kinetic energy and potential energy with the importance of lattice constant.

Sol.

a) The potential energy plot and allowed energy states are as follows:

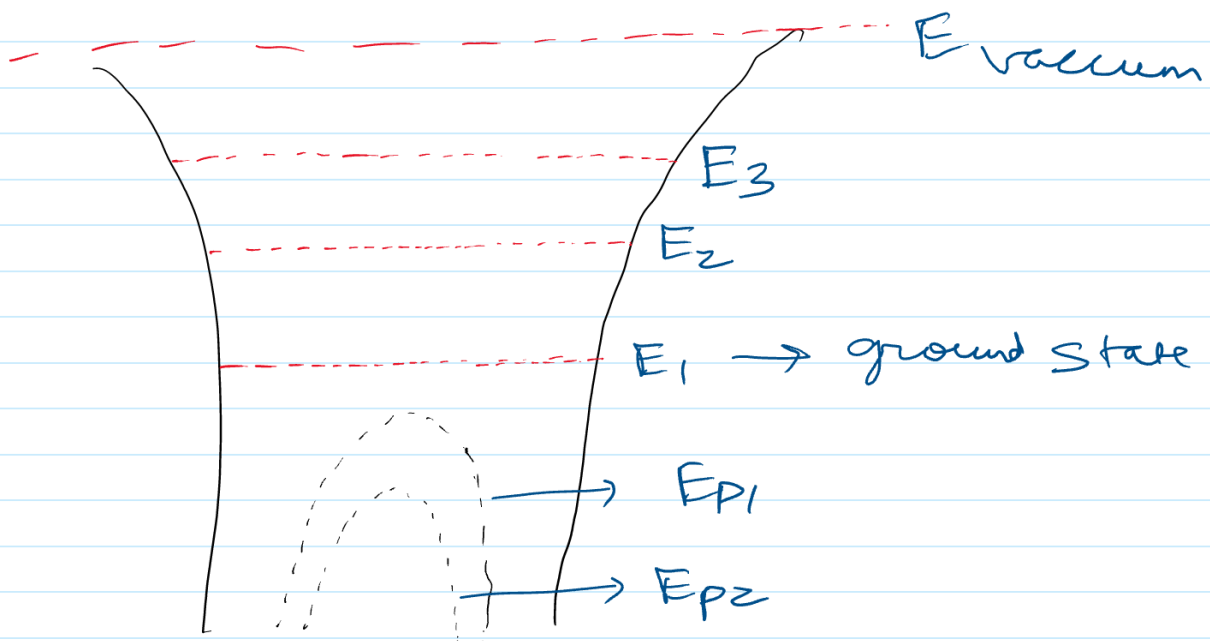


b) Depending on the interatomic distance,
Kinetic Energy $E_k \approx E_l - E_p$.

If $E_k = 0 \rightarrow E_l = E_p$ (Potential energy
= ground state energy)

$\rightarrow$ then a stable covalent bond is
established

$\rightarrow$ This is the condition for covalent bond
formation.



Let us assume E_{p1}, E_{p2} are two peak potential energy levels with $E_{p1} > E_{p2}$.

Clearly we know kinetic Energy $E_k = E_1 - E_p$
where $E_1 \rightarrow$ ground state
 $E_p \rightarrow$ peak potential

In case 1, $E_{k1} = E_1 - E_{p1}$

In case 2, $E_{k2} = E_1 - E_{p2}$

As $E_{p1} > E_{p2} \Rightarrow E_{k1} < E_{k2}$

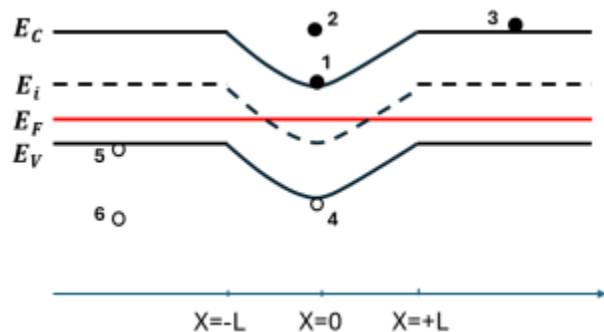
Thus e^- s in the case I has lower kinetic Energy. Lower kinetic energy means lesser speed of e^- within the e^- cloud and stronger covalent bond.

Thus covalent bond weakens in case 2.

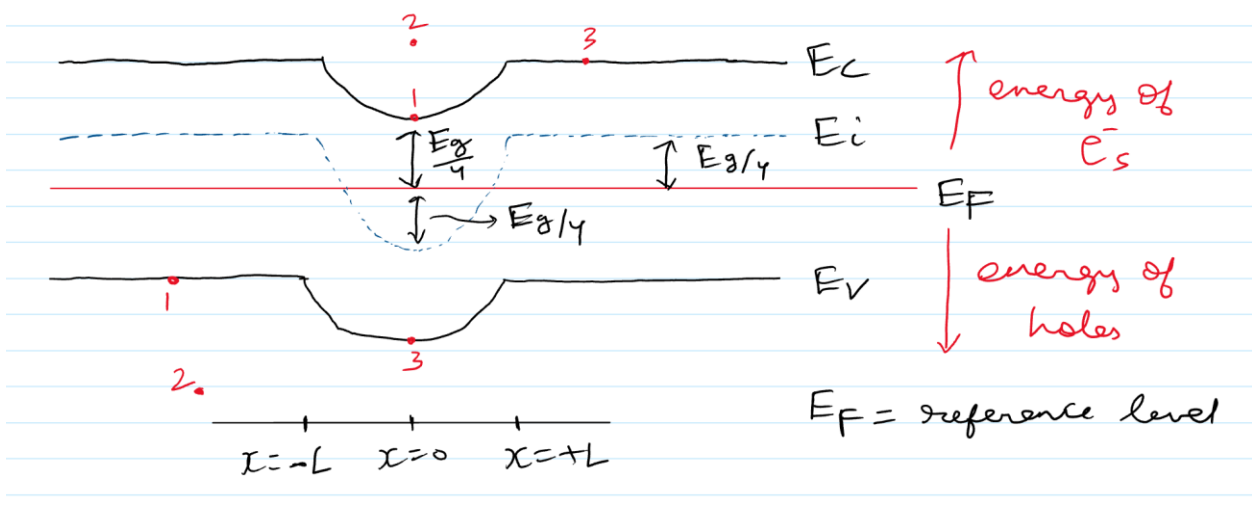
lattice constant relation wrt E_{p1} and E_{p2}

The larger lattice constant (E_{p2}) corresponds to higher potential energy. The total energy of the system decreases with bonding interaction. But we cannot keep decreasing the lattice constant below the equilibrium separation (the stable bond length). If we push atoms very closer (very small lattice constant case) than the equilibrium bond distance then atomic cores repel each other due to strong nuclear and electron-electron repulsion which leads to increase in total energy and eventually to a unstable system.

2) Consider the following energy band diagram. Take the semiconductor represented to be Si maintained at 300 K with $E_i - E_F = E_g/4$ at $x = \pm L$ and $E_F - E_i = E_g/4$ at $x = 0$. Note the choice of E_F as the energy reference level. Find the K.E. and P.E. of the electrons and holes pictured on the diagram.
 Note: No partial marks. Every charge carrier should have correct KE and PE.



Sol.



$$\text{For } \underline{\text{kinetic energy (K.E.)}} = E - E_c$$

↓
energy of location of e^- sitting level

for electron 1, we can visually see that it is sitting across the edge of E_c [or extremely near to E_c only, but let's assume that $e^- 1$ is at E_c energy level only]

$$\text{So } K.E. e^- 1 = E_c - E_c = 0$$

for electron 2, $(K.E.)_2 = E - E_c$

(the amount of band bending basically)
for conduction band

So from reference E_f , we can write

$$(K.E.)_2 = (E_i - E_f)_{x=+L \text{ or } -L} + (E_f - E_i)_{\text{at } x=0} \quad (\text{mathematically})$$

$$(K.E.)_2 = \frac{E_g}{4} + \frac{E_g}{4} = \frac{1.12 \text{ eV}}{2} = 0.56 \text{ eV}$$

for electron 3 $(K.E.)_3 = E - E_c = E_c - E_c = 0$

[$\because e^-3$ is also sitting on edge of E_c so
considering ~~its~~ its energy level as E_c]

For Potential Energy : ~~the~~ energy of holes increases
downwards on the diagram
or energy of e^- s increases upwards on diagram

$$P.E. = E_c - E_f \quad [\because E_c - E_{ref.}] \rightarrow \text{for } e^-s$$

for electron 1 [which is lying on conduction band]

$$(P.E.)_1 = E_c - E_f = (E_c - E_i) - (E_f - E_i)$$

origin
at $x=0$

$$(P.E.)_1 = \frac{E_g}{2} - \frac{E_g}{4} = \frac{E_g}{4} = \frac{1.12}{4} = 0.28 \text{ eV}$$

for electron 2 $P.E|_{2e^-} = E_c - E_F$ [P.E. of $e^- 1$ & $e^- 2$ will be same]
 ↳ at same x -level (origin)

for electron 3

$$[P.E]_{3e^-} = E_c - E_F = (E_c - E_i) + (E_i - E_F) = \frac{E_g}{2} + \frac{E_g}{4}$$

$$= \frac{3E_g}{4} = \frac{3 \times 1.12}{4} = 0.84 \text{ eV}$$

for hole 1 $P.E|_{\text{hole}} = E_V - E_{\text{ref.}} = E_V - E_F$

$$P.E. = (E_V - E_i) - (E_F - E_i) = \frac{E_g}{2} - \frac{E_g}{4} = \frac{E_g}{4} = 0.28 \text{ eV}$$

for hole 3 (at origin $x=0$)

$$P.E. = (E_V - E_i) + (E_i - E_F) = \frac{E_g}{2} + \frac{E_g}{4} = \frac{3E_g}{4} = 0.84 \text{ eV}$$

for hole 2 (sitting far away from V.B.)

but this can affect K.E. but not P.E.

So P.E. of hole 1 & hole 2 will be same [at same x -level]

$$(P.E.)_{\text{hole 1}} = (P.E.)_{\text{hole 2}} = 0.28 \text{ eV}$$

Carrier	K.E (eV)	P..E (eV)
Electron 1	0	0.28
Electron 2	0.56	0.28
Electron 3	0	0.84
Hole 1	0	0.28
Hole 2	0.56	0.28
Hole 3	0	0.84

- 3) a) What is an intrinsic semiconductor?
b) At 0K, are there any free charge carriers?
c) Describe any two methods to increase the charge carriers in the semiconductor? Which method is feasible to increase the free charge carriers?

Sol.

- a) An intrinsic semiconductor is a pure semiconductor without any defect or impurity equal numbers of electrons and holes.
- b) At absolute zero (0 K), all electrons occupy the valence band, the conduction band is empty, as electrons do not have enough energy to jump into the conduction band. So, no free charge carriers
- c) There are two main methods of introducing dopants into semiconductors:
 - 1. **Increasing temperature** — As the temperature rises, more thermal energy is available to electrons. These electrons can gain enough energy to jump from the valence band (VB) to the conduction band (CB), creating more free electrons and holes. This increases the carrier concentration.

2. **Adding impurities (Doping)** — Introducing controlled amounts of impurity atoms (dopants) into a pure semiconductor dramatically increases **carrier concentration** by creating additional energy levels within the bandgap.

- In **n-type doping**, donor impurities (like phosphorus in silicon) contribute extra electrons that are easily excited into the conduction band, increasing electron concentration.
- In **p-type doping**, acceptor impurities (like boron in silicon) create holes in the valence band, increasing hole concentration.

Doping **is more feasible** because it offers better accuracy, uniformity, and control.